

Constant-Rate Dual-Railing of Bounded Polynomial GPACs: UCNC 2025 Problem 1, General n -Dimensional Case

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Abstract

The selective dual-railing construction of Haisler, Huang, Migunov, Mohammed, and Provence (UCNC 2025) converts a polynomial vector field $\dot{y}_i = p_i(y)$ into a chemical reaction network on $2n$ non-negative species $(u_1, v_1, \dots, u_n, v_n)$ with $y_i = u_i - v_i$. A shared annihilation term $-k u_i v_i$ prevents each rail-pair from drifting apart. UCNC 2025 Problem 1 asks: given a bounded polynomial GPAC trajectory, does there exist a single constant rate k (independent of the bound) for which the dual-railed system remains bounded on $[0, \infty)$? We answer affirmatively for the general n -dimensional case starting from the dual-rail split of the GPAC initial condition. Concretely: for every polynomial vector field $p: \mathbb{R}^n \rightarrow \mathbb{R}^n$ admitting a bounded solution y_{Sol} with $\|y_{\text{Sol}}(t)\|_\infty \leq \beta$, the rate

$$k_\star := \max_i \frac{2M_i(\beta)}{3\beta^2}, \quad M_i(\beta) := \max_{w \in [0, 2\beta]^{2n}} (\widehat{p}_i^+(w) + \widehat{p}_i^-(w)),$$

suffices: any rational $k > k_\star$ keeps the dual-rail trajectory inside the box $\{(u, v) \geq 0 : u_i + v_i \leq 2\beta\}$, and the difference $u_i - v_i$ recovers $y_{\text{Sol}, i}$ exactly by ODE uniqueness. The construction has been formalized in Lean 4 with no unproved goals and no axioms beyond Lean’s standard logical core (propositional extensionality, the axiom of choice, and the soundness of quotient types).

1 Setup

Polynomial GPAC. Fix $n \geq 1$ and a tuple $p = (p_1, \dots, p_n) \in \mathbb{Q}[X_1, \dots, X_n]^n$. We write $\widehat{p}_i^+, \widehat{p}_i^- \in \mathbb{Q}[U_1, V_1, \dots, U_n, V_n]$ for the positive and negative parts of the dual-rail substitution $y_j \mapsto U_j - V_j$ in p_i : every monomial of p_i with positive coefficient lands in $\widehat{p}_i^+$ after binomial expansion, every monomial with negative coefficient lands (with sign flipped) in $\widehat{p}_i^-$. Both have non-negative coefficients. The fundamental identity

$$\widehat{p}_i^+(u, v) - \widehat{p}_i^-(u, v) = p_i(u - v), \tag{1}$$

holds in $\mathbb{Q}[U, V]$ and is the algebraic engine of the construction.

Constant-rate dual-rail system. For a positive rational k , the *constant-rate dual-rail GPAC associated to (p, k)* is the polynomial vector field on $\mathbb{R}^{2n}$:

$$\begin{cases} \dot{u}_i = \widehat{p}_i^+(u, v) - k u_i v_i, \\ \dot{v}_i = \widehat{p}_i^-(u, v) - k u_i v_i, \end{cases} \quad i = 1, \dots, n. \tag{2}$$

With non-negative coefficients and a single per-pair degradation, this is mass-action implementable as a CRN.

Bounded GPAC. A *bounded GPAC datum* is a tuple $(p, y_0, y_{\text{Sol}}, \beta)$ with $\beta > 0$, $y_0 \in \mathbb{Q}^n$, and $y_{\text{Sol}}: \mathbb{R} \rightarrow \mathbb{R}^n$ such that $y_{\text{Sol}}(0) = y_0$ (cast to reals), $\dot{y}_{\text{Sol},i}(t) = p_i(y_{\text{Sol}}(t))$ for $t \geq 0$, and $|y_{\text{Sol},i}(t)| \leq \beta$ for $t \geq 0$ and every i . In particular $|(y_0)_i| \leq \beta$.

Dual-rail split of the initial condition. Define $u_0, v_0 \in \mathbb{Q}_{\geq 0}^n$ by

$$(u_0)_i := \max((y_0)_i, 0), \quad (v_0)_i := \max(-(y_0)_i, 0),$$

so $(u_0)_i - (v_0)_i = (y_0)_i$ and $(u_0)_i + (v_0)_i = |(y_0)_i| \leq \beta$.

Box. The forward-invariant set we will exhibit is

$$B_\beta := \{w \in \mathbb{R}_{\geq 0}^{2n} : w_{2i} + w_{2i+1} \leq 2\beta \ \forall i\}.$$

We index the $2n$ dual-rail coordinates by $K \in \{0, \dots, 2n-1\}$ with $u_i = w_{2i}$ and $v_i = w_{2i+1}$.

2 Main theorem

Theorem 1 (UCNC 2025 Problem 1, general n). *Let $(p, y_0, y_{\text{Sol}}, \beta)$ be a bounded polynomial GPAC datum and let*

$$k_\star := \max_{i \in \{1, \dots, n\}} \frac{2M_i(\beta)}{3\beta^2}, \quad M_i(\beta) := \sup_{w \in [0, 2\beta]^{2n}} (\hat{p}_i^+(w) + \hat{p}_i^-(w)).$$

For every rational k with $k > k_\star$, there exists a global solution $\hat{y}: [0, \infty) \rightarrow \mathbb{R}^{2n}$ of (??) with $\hat{y}(0) = (u_0, v_0)$ such that, for every $t \geq 0$,

- (i) **Non-negativity.** $\hat{y}(t)_K \geq 0$ for every K .
- (ii) **σ -bound.** $\hat{y}(t)_{2i} + \hat{y}(t)_{2i+1} \leq 2\beta$ for every i .
- (iii) **Dual-rail identity.** $\hat{y}(t)_{2i} - \hat{y}(t)_{2i+1} = y_{\text{Sol},i}(t)$ for every i .

The bound (ii) combined with (i) yields the box invariance $\hat{y}(t) \in B_\beta$, and (iii) closes the dual-rail implementation: $\hat{y}_{2i} - \hat{y}_{2i+1}$ tracks the original GPAC trajectory exactly, so the CRN is a faithful simulator of y_{Sol} .

The proof decomposes into four sub-lemmas, each formally stated in the Lean development. Sections ??–?? walk through them.

3 Algebraic identities

Drift difference. Subtracting the two lines of (??) and using (??):

$$\dot{u}_i - \dot{v}_i = \hat{p}_i^+(u, v) - \hat{p}_i^-(u, v) = p_i(u - v). \quad (3)$$

The annihilation term cancels. Hence the difference $z_i := u_i - v_i$ satisfies the original GPAC.

Drift sum on the box. Adding the two lines of (??):

$$\dot{u}_i + \dot{v}_i = \widehat{p}_i^+(u, v) + \widehat{p}_i^-(u, v) - 2k u_i v_i.$$

Since both $\widehat{p}_i^+$ and $\widehat{p}_i^-$ have non-negative coefficients, the absolute polynomial $|p|_i := \widehat{p}_i^+ + \widehat{p}_i^-$ is monotonic in each coordinate over $\mathbb{R}_{\geq 0}^{2n}$. On the box $[0, 2\beta]^{2n}$, $|p|_i$ is therefore bounded by $M_i(\beta)$ (its supremum on the box). Hence on B_β we have the upper bound

$$\dot{u}_i + \dot{v}_i \leq M_i(\beta) - 2k u_i v_i. \quad (4)$$

Face inequality. A short algebraic computation shows that on the upper face $u_i + v_i = 2\beta$ subject to $|u_i - v_i| \leq \beta$, the product $u_i v_i$ is bounded below:

$$4u_i v_i = (u_i + v_i)^2 - (u_i - v_i)^2 \geq (2\beta)^2 - \beta^2 = 3\beta^2. \quad (5)$$

Strict inward-pointing at the upper face. Combining (??) and (??), on the face $u_i + v_i = 2\beta$ inside B_β with the additional dual-rail constraint $|u_i - v_i| \leq \beta$,

$$\dot{u}_i + \dot{v}_i \leq M_i(\beta) - 2k \cdot \frac{3}{4}\beta^2 = M_i(\beta) - \frac{3}{2}k\beta^2.$$

This is strictly negative whenever $k > 2M_i(\beta)/(3\beta^2)$, which is exactly the threshold condition $k > k_*$. Hence:

Lemma 2 (upper-face strict negativity). *If $k > k_*$, $\beta > 0$, $w \in B_\beta$ with $w_{2i} + w_{2i+1} = 2\beta$ and $|w_{2i} - w_{2i+1}| \leq \beta$, then the σ -drift $\dot{u}_i + \dot{v}_i$ at w is strictly less than zero.*

4 Picard existence

The right-hand side of (??) is polynomial, hence locally Lipschitz on $\mathbb{R}^{2n}$. By the standard Picard–Lindelöf argument, combined with the a priori bound $\widehat{y}(t) \in B_\beta$ established in the next section, there exists a global solution $\widehat{y}: [0, \infty) \rightarrow \mathbb{R}^{2n}$ of (??) with $\widehat{y}(0) = (u_0, v_0)$.

5 The invariant: nonneg + diff + σ -bound

We prove the three-pronged invariant on every interval $[0, T)$ on which the candidate solution exists. The three prongs — non-negativity, difference bound, and σ -bound — are proved in this order and assembled in Section ??.

5.1 Componentwise non-negativity

Lemma 3. *Let $\widehat{y}: \mathbb{R} \rightarrow \mathbb{R}^{2n}$ satisfy (??) on $[0, T)$ with non-negative initial condition. Then $\widehat{y}(t)_K \geq 0$ for all $t \in [0, T)$ and all K .*

Proof. The system (??) admits a CRN decomposition: each component $\dot{w}_K = (\text{production}) - k w_K w_{K'}$ where the production term is $\widehat{p}_i^+$ or $\widehat{p}_i^-$ (both non-negative on $\mathbb{R}_{\geq 0}^{2n}$), and the degradation $-k w_K w_{K'}$ is the product of w_K with another component. At any face $w_K = 0$ the production is non-negative and the degradation vanishes, so the face is inward-pointing. The standard local non-negativity argument for CRNs then yields $\widehat{y}(t) \geq 0$ for all t in the lifetime of the solution. $\square$

5.2 Difference bound by ODE uniqueness

Lemma 4. *Under the hypotheses of Theorem ??, on $[0, T)$,*

$$|\widehat{y}(t)_{2i} - \widehat{y}(t)_{2i+1}| \leq \beta.$$

Proof. Let $z_i(t) := \widehat{y}(t)_{2i} - \widehat{y}(t)_{2i+1}$. By (??), $\dot{z} = p(z)$ on $[0, T)$. At $t = 0$:

$$z_i(0) = (u_0)_i - (v_0)_i = \max((y_0)_i, 0) - \max(-(y_0)_i, 0) = (y_0)_i = y_{\text{Sol},i}(0).$$

Both z and y_{Sol} satisfy the original polynomial GPAC $\dot{w} = p(w)$ with the same initial condition. On any compact sub-interval $[0, t] \subset [0, T)$, both solutions are bounded (continuity on a compact interval) and p is locally Lipschitz, so the standard ODE uniqueness theorem on a compact interval gives $z = y_{\text{Sol}}$ on $[0, t]$. Letting $t \uparrow T$: $z(t) = y_{\text{Sol}}(t)$ on $[0, T)$. By the bound assumption, $|y_{\text{Sol},i}(t)| \leq \beta$. $\square$

Remark 5. *This step does not use the σ -bound and so can be carried out independently. Critically, this is what breaks a potential circular dependency: the Nagumo σ -barrier needs the diff bound inside the strict-negativity lemma ??, but the diff bound itself uses only ODE uniqueness on compact subintervals (with sup-norm bounds extracted by compactness, not by B_β).*

5.3 The joint σ -Nagumo barrier

This is the analytic core. Define $\sigma_i(t) := \widehat{y}(t)_{2i} + \widehat{y}(t)_{2i+1}$.

Lemma 6. *Under the hypotheses of Theorem ?? and assuming $k > k_\star$, on $[0, T)$,*

$$\sigma_i(t) \leq 2\beta \quad \forall i.$$

Proof. By Lemmas ?? and ??, $\widehat{y}_K \geq 0$ and $|z_i| \leq \beta$ on $[0, T)$. Suppose for contradiction there exist $t' \in [0, T)$ and i_0 with $\sigma_{i_0}(t') > 2\beta$. Define

$$S := \{\tau \in [0, t'] : \sigma_i(\tau) \leq 2\beta \forall i\}.$$

$0 \in S$. At $t = 0$, $\sigma_i(0) = (u_0)_i + (v_0)_i = |(y_0)_i| \leq \beta < 2\beta$.

Define $\tau_\star := \sup S \in [0, t']$. By sequential continuity of each σ_i , $\tau_\star \in S$, hence $\sigma_i(\tau_\star) \leq 2\beta$ for every i .

$\tau_\star < t'$. By assumption $\sigma_{i_0}(t') > 2\beta$, so $t' \notin S$ and $\tau_\star < t'$.

Box property at $\tau_\star$. Since $\widehat{y}_K(\tau_\star) \geq 0$ and $\sigma_i(\tau_\star) \leq 2\beta$, each component $\widehat{y}_K(\tau_\star)$ is bounded above by its σ -partner's sum minus the other component, which is at most $\sigma_i(\tau_\star) \leq 2\beta$. Hence $\widehat{y}(\tau_\star) \in B_\beta$.

Per-coordinate buffers. For each i :

- If $\sigma_i(\tau_\star) < 2\beta$: by continuity of σ_i , there exists $\delta_i > 0$ such that $\sigma_i(\tau) \leq 2\beta$ on $[\tau_\star, \tau_\star + \delta_i]$.
- If $\sigma_i(\tau_\star) = 2\beta$: by Lemma ?? (applied with $w = \widehat{y}(\tau_\star)$, using the box property, the face condition, and Lemma ??), $\dot{\sigma}_i(\tau_\star) < 0$. A function with strictly negative derivative at a point lies below its value on a right neighborhood: explicitly, by the definition of the derivative, for some $\delta_i > 0$ and all $\tau \in [\tau_\star, \tau_\star + \delta_i]$,

$$\sigma_i(\tau) \leq \sigma_i(\tau_\star) + \frac{1}{2} \dot{\sigma}_i(\tau_\star) (\tau - \tau_\star) \leq \sigma_i(\tau_\star) = 2\beta.$$

Uniform buffer. Take $\delta := \min(\min_i \delta_i, t' - \tau_\star) > 0$. The minimum over i is well-defined and positive because $\{1, \dots, n\}$ is finite and every $\delta_i > 0$. Then on $[\tau_\star, \tau_\star + \delta]$, all $\sigma_i \leq 2\beta$, hence $\tau_\star + \delta \in S$.

Contradiction. But $\tau_\star + \delta > \tau_\star = \sup S$, contradicting the definition of supremum. $\square$

Remark 7 (Why the barrier is joint). *A naive per-coordinate barrier — defining $\tau_i := \sup\{\tau : \sigma_i(\tau) \leq 2\beta\}$ separately for each i — fails because Lemma ??’s box hypothesis $w \in B_\beta$ requires all coordinates to satisfy $w_{2j} + w_{2j+1} \leq 2\beta$, not just the i -th pair. Choosing $i_\star$ as the last to cross does not help, because by that time other σ_j may already have escaped. The joint sup $\tau_\star := \sup\{\tau : \forall i, \sigma_i(\tau) \leq 2\beta\}$ is the correct barrier: at $\tau_\star$ all σ_i are simultaneously $\leq 2\beta$, hence $w(\tau_\star) \in B_\beta$, hence the strict inward-pointing condition applies to every i saturating the upper face.*

6 Closing the dual-rail identity

Lemma 8. *Under the hypotheses of Theorem ??, on $[0, \infty)$,*

$$\widehat{y}(t)_{2i} - \widehat{y}(t)_{2i+1} = y_{\text{Sol},i}(t).$$

This is what we already proved as Lemma ??, but now over the global interval $[0, \infty)$ (not just on a finite $[0, T)$).

Proof. The argument of Lemma ?? applies verbatim with $T = \infty$, since the global Picard solution $\widehat{y}$ exists on $[0, \infty)$ and ODE uniqueness on every compact $[0, t]$ pins the difference to y_{Sol} . $\square$

7 Assembly

Theorem ?? now follows. Pick a rational $k > k_\star$ (possible by density of $\mathbb{Q}$ in $\mathbb{R}$); Lemma ?? gives clause (i), Lemma ?? gives clause (ii), and Lemma ?? gives clause (iii).

Remark 9 (Boundary case $n = 0$). *The $n = 0$ case is vacuous: with no variables, all universal statements are trivially satisfied.*

8 Comparison with the scalar cubic

For $p(y) = 1 - y^3$ on $[-1, 1]$, the threshold $k_\star = 6$ matches the saddle-node value derived in the companion scalar-cubic note via direct phase-plane analysis. The general construction here gives the same constant (up to the rounding implicit in $\mathbb{Q}$ -density), but the proof is substantially more abstract: it does not exploit the explicit 1-dim structure, instead leveraging only the σ -Nagumo argument. The advantage is generality; the cost is a slightly weaker statement (the box width 2β vs the sharper scalar-cubic invariant).

Notes

The full Lean 4 formalization, organized along the lemma sequence above (algebraic identities, threshold witness, Picard existence, the three-pronged invariant, dual-rail identity, main theorem), is available in the Ripple repository under `Ripple/DualRail/ConstantAnnihilationGeneral.lean`. The proof depends only on Lean’s standard axiomatic core (propositional extensionality, choice, soundness of quotients).

The companion note on the scalar cubic case ($p(y) = 1 - y^3$, threshold $k_\star = 6$) is at `notes/
scalar-cubic-dual-rail.tex`.